$\boxed{№1.4}$ $\boxed{a \geq 0}$

a) $T(n) = \begin{cases} 1, & n \leq a \\ T(n-a)+1 \end{cases}$

Оскільки $T(n) = T(n-a)+1 =$
$= T(n-2a)+2 = \dots \Rightarrow \forall k \in \mathbb{N}$:
$T(n) = T(n-ka)+k$, $n-ka \leq a$
$\Rightarrow \underline{\underline{T(n) = \left[\frac{n}{a}\right]+1}}$.

b) $T(n) = \begin{cases} 1, & n=0 \\ T(n-1)+2^n, & n \geq 1 \end{cases}$

$T(n) = T(n-1)+2^n = T(n-2)+2^{n-1}+2^n =$
$\dots = T(0)+\underbrace{2^1+2^2+\dots+2^n}_{\text{ГП}} = 1+$
$+2^{n+1}-2 = \underline{\underline{2^{n+1}-1}}$

c) $T(n) = \begin{cases} 1, & n=1 \\ 2T(\lceil \frac{n}{2} \rceil)+1, & n \geq 2 \end{cases}$

$k = \lceil \log_2 n \rceil \Rightarrow T(2^k) = 2T(2^{k-1})-1 =$
$= \dots = 2^k T(1)+2^{k-1}+\dots+2^0 = 2^{k+1}-1$
$\Rightarrow \underline{\underline{T(n) = 2^{\lceil \log_2 n \rceil+1}-1}}$